Does an attention head look at a country, or just at a position?

Experiment 4, Figures 1 and 2: the design from first principles, and how each figure is computed. GPT-2 small, attention from the final token.

Question. Exp 9–10 maps showed that L11H8's most-attended key became a country only when the country sat right before the final token. Either the head is **positional** (reads position $T-1$ whatever sits there) or **content-driven** (reacts to the country itself, perhaps mostly nearby). The earlier maps cannot separate the two; a matched with-and-without comparison can.

- **Measure.** At the final position each of the 144 heads (12 layers $\times$ 12) spreads one unit of attention over the earlier positions, BOS included. $a_h(\tau)$ is head h's share on position τ, so a change of 0.065 means 6.5 points of its attention moved onto τ.
- **Vary only the token at** τ, at every distance $d = T - \tau$ from 1 (adjacent) to 31. A positional head should not care what sits there ($\Delta \approx 0$); a country-tracking head should react wherever the country goes.
- **Two baselines.** The row's own random token at τ, and a placebo, a capitalised non-country word from the same token-id class, to separate "a country" from "a word-like token".

Data. 200 random rows per filler type: BOS plus 32 random tokens, no repeats, final token $T{=}32$ never edited. *Common* filler = token ids 256–999, *rare* = 1000–39999. For every τ and each of 11 countries, three sequences differ *only* at τ (asserted): the original, the country, the placebo. About 68 000 country and 68 000 placebo sequences per filler type.

original	BOS	she	sc	… 26 random tokens …	age	dis	ious
country	BOS	she	sc	… 26 random tokens …	France	dis	ious
placebo	BOS	she	sc	… 26 random tokens …	matched word	dis	ious

(τ = 30, d = 2; orange = final token, the query)

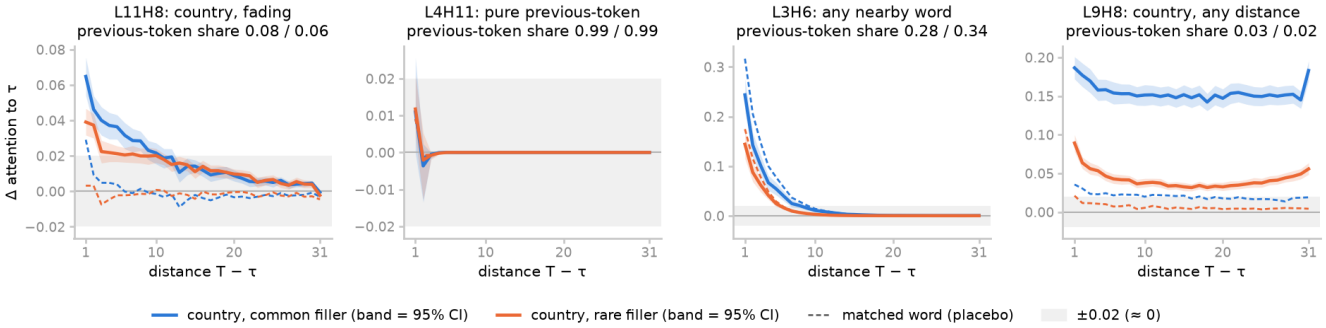
The number behind every curve and cell, with $\tau = 32 - d$ and a_h the attention from the final token to τ:

$$\Delta_h(d) \;=\; \frac{1}{200}\sum_{\text{rows}}\;\frac{1}{11}\sum_{\text{countries } c}\Big[\,a_h(\text{row with } c \text{ at } \tau)\;-\;a_h(\text{row})\,\Big]$$

The row is the unit (its 11 countries are averaged first). Bands are 95% bootstrap CIs over rows. A cell is *meaningful* when a Wilcoxon test over rows gives BH-adjusted $q \le 0.05$ (over all 31 distances $\times$ 144 heads) and $|\Delta| \ge 0.02$; these thresholds were fixed before the data.

Figure 1 · One head, one curve

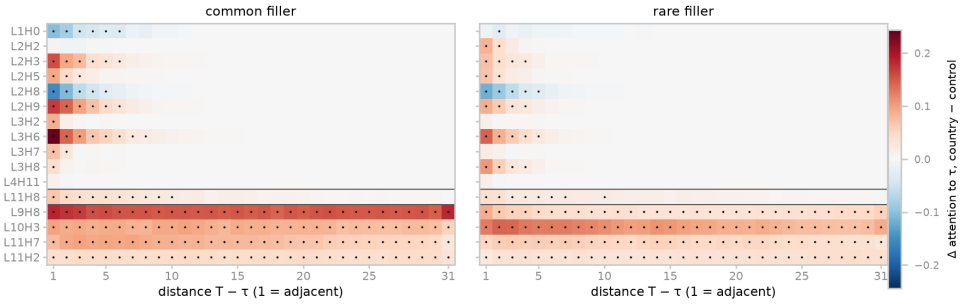
x = distance $T - \tau$, y = $\Delta_h(d)$



Built as: one panel per head; solid = the formula above; band = 95% CI over rows; dashed = the same with the placebo (mean only); grey band = $|\Delta| < 0.02$, treated as zero; "previous-token share" = the head's attention to position 31 in the unmodified rows (common / rare). Each panel has its own y-axis, and the heads are hand-picked examples. **Read:** L4H11 moves at $d{=}1$ only (position-only). L3H6 follows any capitalised word, nearby only. L11H8 follows countries above the placebo, fading with distance. L9H8 follows countries at every distance.

Figure 2 · Many heads at once

colour = $\Delta_h(d)$, dots = meaningful cells



Built as: the same $\Delta_h(d)$, country minus control, as colour: row = head, column = distance, one panel per filler type; red = attention to τ rises, blue = it falls, one symmetric scale for both panels. Rows: Exp 1's 11 previous-token heads, L11H8, four late heads. **Read:** a blank row ignores the token at τ (L4H11). L2H3, L2H9 and L3H6 are red within about 8 tokens; L1H0 and L2H8 are blue; the four late heads are red at all 31 distances. There is no placebo here, so red means something at τ pulls attention, not necessarily a country. Only Figure 1 separates the two.

Limits. Δ is a change in where a head looks, not in what it writes. Placebo words are not matched on token frequency, and the control is a random token, so country − control mixes "a country" with "not a random filler token". 11 countries, 200 rows per filler type, one model.